ROYYA CHAKRAVARTHI

Digital Systems Engineer | RTL Design | Full Stack Developer

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PROFESSIONAL SUMMARY

M.E. Digital Systems candidate at Osmania University with dual expertise in hardware engineering and software systems. Proficient in synthesizable RTL design using Verilog HDL, Cadence Xcelium simulation, and INT8 quantization for neural accelerators, alongside full-stack enterprise development with Java, Spring Boot, React.js, and Node.js. Experienced in hardware verification, RF sub-system testing, and scalable cloud application architecture.

EDUCATION

University College of Engineering, Osmania University — Hyderabad, India
Master of Engineering (M.E.) in Digital Systems Pursuing (Expected Aug 2027)

Anurag University — Hyderabad, India
Bachelor of Technology (B.Tech) in Electronics & Communication Engineering 2020 – 2024 | CGPA: 8.58 / 10

TECHNICAL SKILLS

Digital Systems & VLSI: Verilog HDL, RTL Design, Cadence Xcelium, SimVision, FPGA Concepts, INT8 Quantization, ALU Datapaths

Software Development: Java (Core & J2EE), Spring Boot, React.js, Node.js, Express.js, MongoDB, MySQL, JavaScript (ES6+), C

Tools & Platforms: Git, GitHub, Linux, Docker (Basics), Postman, RESTful APIs, Eclipse IDE, MATLAB

EXPERIENCE

Lanvit Technologies — Hyderabad, India
Java Full Stack Developer Intern June 2025 – September 2025 [Internship]
• Developed responsive full-stack modules using Java, Spring Boot, and React.js, improving client data processing speed.
• Designed secure RESTful API endpoints and connected backend microservices with relational MySQL databases.

Inspanner Academy — Hyderabad, India
Java Full Stack Trainee October 2024 – April 2025 [Professional Training]
• Completed intensive training in Core Java, Spring Boot microservices, Hibernate ORM, and modern React.js frontend architecture.

Astra Microwave Products Limited — Hyderabad, India
Project Trainee February 2024 – June 2024 [Project Trainee]
• Assisted in testing and diagnostic analysis of high-frequency RF and microwave sub-systems using Spectrum Analyzers and Power Meters.

KEY PROJECTS

Hardware Accelerator Design of LeNet-5 CNN | Verilog HDL, Cadence Xcelium, Python
• Designed custom synthesizable Verilog accelerator for LeNet-5 CNN; applied INT8 quantization reducing memory footprint by 75%.
• Verified cycle-accurate RTL execution and line-buffer sliding window convolution against software tensors in Cadence Xcelium.

Harvixa (Rice Mill Express Platform) | React.js, Node.js, Express, MongoDB, Razorpay
• Architected full-stack grain logistics ERP automating moisture-adjusted calculations, inventory management, and digital payments.

RTL 32-Bit ALU Design & Verification | Verilog HDL, Cadence SimVision
• Designed 32-bit ALU supporting 16 arithmetic, logic, and barrel-shift operations with zero/overflow flags verified via testbenches.